Superstore Retail Sales: Executive Summary

Exploratory Data Analysis | Datheon internship assignment | Orders Jan 2015 to Dec 2018 | 9,993 cleaned order lines

Total Sales	Total Profit	Profit Margin	Orders
$2.30M	**$286.4K**	**12.5%**	**5,009**
Avg Order Value	Customers	Sales Growth 2018	Loss-making lines
$459	**793**	**+20.4%**	**18.7%**

Headline

The business is growing and profitable, and demand follows a stable year-end peak. The main weakness is discounting: lines discounted above 20% lost $135.4K in total, and the weakest areas (Furniture Tables and Bookcases, the Central region, Texas) are where discounts are heaviest. Fixing discount discipline is the fastest route to higher profit.

Key insights

1. Discounts above 20% are almost always loss-making

What the data shows: 1,392 lines (13.9% of all lines) are discounted above 20%. Together they lost $135.4K, and 97% of them individually lost money. Lines with no discount earn a 29.5% margin versus 11.9% for 1-20% discounts. The Spearman correlation between discount and profit is -0.54.
Why it matters: Deep discounts and profit move in opposite directions, and every discount band above 20% has a negative total profit. This is an association, not proof of cause, because discounted lines also differ in product mix and region.
Action: Set a 20% standard discount ceiling, require manager approval above it, and track margin after discount every month. Pilot the rule in one region first to measure the real effect.

2. Furniture sells a lot but earns very little

What the data shows: Furniture is 32.3% of sales but only 6.4% of profit (margin 2.5% vs 17.4% for Technology and 17.0% for Office Supplies). Tables lost $17.7K (-8.6% margin) and Bookcases lost $3.5K. Table lines discounted above 20% lost $30.7K, while Table lines at 20% or less made $13.0K.
Why it matters: The category's weak profit is concentrated in two sub-categories, and heavy discounting appears to be a major part of the Tables problem. Chairs and Furnishings remain profitable.
Action: Review prices and supplier costs for Tables and Bookcases, stop deep discounts on them, and move promotional budget toward Technology and Office Supplies.

3. Sales are concentrated in September to December

What the data shows: September to December generate 51.6% of sales (November alone 15.3%). February is the weakest month (2.6% of sales). September, November and December were the three biggest months in 4 of 4 years.
Why it matters: The pattern is stable, so inventory, staffing and marketing can be planned in advance. The likely cause is year-end buying, although the data cannot confirm the reason.
Action: Build stock and shipping capacity before September. In January-February use bundles and value offers rather than deep discounts to lift the slow months.

4. The Central region has the weakest margin

What the data shows: Central has $501.2K in sales (21.8%) but a 7.9% margin versus 14.9% in West. Its average discount is 24.0% versus 10.9% in West, and 27.7% of Central lines are discounted above 20% (West: 3.7%). Texas lost $25.7K with an average discount of 37.0%. Inside Central, Indiana (average discount 0.0%) earned a 34.3% margin, while Illinois (average discount 39.0%) ran at -15.7%.
Why it matters: Central's lower margin coincides with much heavier discounting. Product mix and local costs may also contribute, so a drill-down is needed before drawing a final conclusion.
Action: Audit discount approvals in Central, starting with Texas. Compare Central and West discounting product by product, and set a minimum margin target for each region.

5. Sales growth is strong, but margin slipped in 2018

What the data shows: Sales grew 29.5% in 2017 and 20.4% in 2018 (to $733.2K), but 2018 profit grew only 14.2% and the margin fell from 13.4% to 12.7%.
Why it matters: Each extra dollar of sales is now worth slightly less profit, so growth is becoming a little less efficient.
Action: Make profit margin a headline KPI next to sales, and link sales targets to margin, not revenue alone.

6. The customer base is broad, and the biggest customer by sales lost money

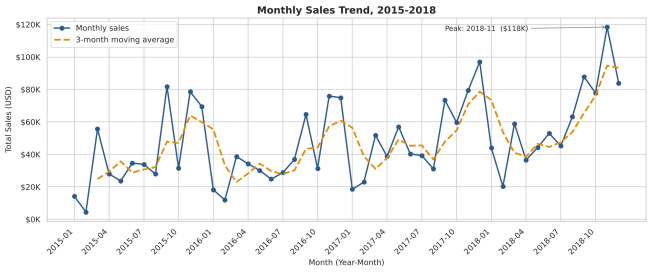
What the data shows: There are 793 customers. The top 10 contribute only 6.7% of sales and the top 20% contribute 48.1%. The #1 customer by sales (Sean Miller) generated $25.0K in sales but -$2.0K in profit. 155 customers (19.5%) have negative total profit, adding up to -$71.2K.
Why it matters: Concentration risk is low, but ranking customers by sales alone can reward accounts that cost money.

Action: Show profit next to sales in customer reports, review the discounts given to loss-making accounts, and use loyalty offers to grow the broad middle tier.

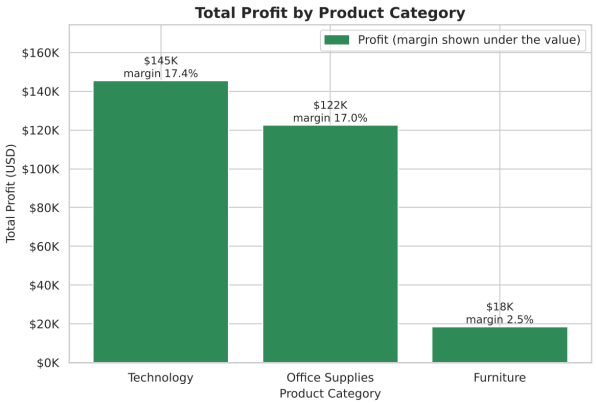
7. Returned orders take a noticeable share of sales

What the data shows: 296 orders (5.9% of all orders) appear in the appended Returns table, worth $180.5K (7.9% of sales). Their average order value is $610 versus $449 for other orders. Technology has the highest share of its sales returned (8.7%).
Why it matters: Returned orders are worth more than a typical order, so each return costs more than the order count suggests. Return reasons and refund amounts are not in the data, so the exact profit impact cannot be measured here.
Action: Start recording return reasons and refund values, and investigate Technology returns first.

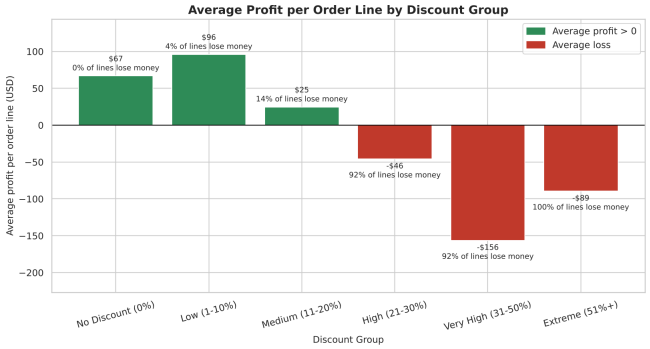
Key charts



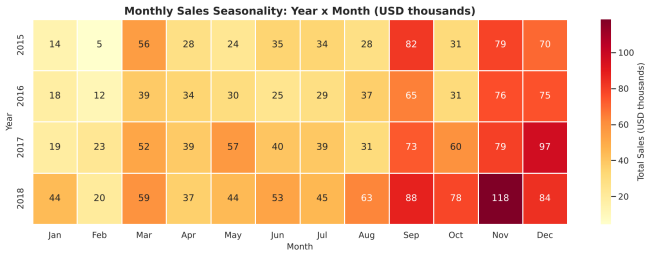
Monthly sales trend



Profit by category



Average profit by discount group



Seasonality heatmap

Recommendations (priority order)

- **Cap standard discounts at 20%** and require approval above it.
- **Audit the Central region**, starting with Texas.
- **Reprice or rework Tables and Bookcases**; stop deep discounts on them.
- **Plan for the Sep-Dec peak**; use bundles, not deep discounts, in Jan-Feb.
- **Report margin and customer profit**, not only sales.
- **Capture return reasons and refund values**.
- **Fix the data export**: one table per file.

Data notes

The uploaded file contained 806 stray rows from appended People and Returns tables; they were set aside (the returned Order IDs were reused). 1 duplicate sales line was removed and 11 missing Postal Codes were filled. Outliers were investigated and kept because they are genuine. Discount and profit are associated, but causation is not proven.